

Research Plan:

The Effect of Background Music Tempo on Typing Speed

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PS50008A: Research Methods & Experimental Design

Goldsmiths, University of London

Spring 2026

Word Count: 802

AI Use Declaration

I confirm that this work is my own. No AI tools were used in the preparation of this submission.

Typing speed is an increasingly valuable workplace skill, with research suggesting that efficient keyboard use is associated with reduced occupational frustration and improved job satisfaction (Qwerty, 2025).

The present study investigates whether background music tempo influences typing performance. It is hypothesised that participants will type significantly faster (more words per minute) when listening to fast-tempo music compared to slow-tempo music. This is a one-tailed (directional) hypothesis. A directional prediction is appropriate because previous research on arousal and performance suggests that faster tempos increase physiological arousal, which may facilitate faster motor responses in simple tasks like typing.

Method

Design

An independent groups (between-participants) design was used. The independent variable was background music tempo, with two levels: fast tempo (140 beats per minute) and slow tempo (60 beats per minute). The dependent variable was typing speed, operationalised as the number of words correctly typed per minute.

An independent groups design was chosen rather than a repeated measures design for two reasons. First, a repeated measures design would introduce order effects—participants might improve through practice or fatigue over two typing sessions, confounding the effect of music tempo. Second, participants might guess the hypothesis if exposed to both conditions, potentially altering their behaviour (demand characteristics). Random allocation to conditions controls for individual differences in baseline typing ability.

Participants

Forty participants (20 per condition) will be recruited via opportunity sampling from the undergraduate population at Goldsmiths, University of London. Participants will be aged 18-30 years to reduce variability in typing experience associated with generational differences in computer use. An approximately equal distribution of male and female participants will be recruited across conditions. Participants will be randomly allocated to conditions using a random number generator.

Exclusion criteria include any diagnosed hearing impairment; motor impairment affecting hand or finger movement; professional typists or those with formal typing training; and non-native English speakers (as the typing task involves English text). Participants will be naive to the experimental hypothesis and informed only that the study investigates factors affecting typing performance.

Materials

Two instrumental music tracks will be used, matched on all characteristics except tempo. The fast tempo condition uses a 140 BPM electronic instrumental track; the slow tempo condition uses a 60 BPM electronic instrumental track. Both tracks are instrumental only (no lyrics), in a major key, at moderate volume (65 dB), and five minutes in duration. Tracks will be sourced from royalty-free music libraries and pilot-tested to ensure equivalent perceived pleasantness.

The typing task consists of a standardised 300-word passage displayed on screen, containing common English words with no technical jargon, emotionally neutral content, displayed in black 14pt Arial font on a white background. Equipment includes a standard

desktop computer with QWERTY keyboard, over-ear headphones for music delivery, typing software that records words per minute and errors, stopwatch, consent forms, and debrief sheets.

Procedure

Participants will be tested individually in a quiet laboratory room. Upon arrival, participants will read an information sheet and sign a consent form. They will be randomly allocated to either the fast-tempo or slow-tempo condition.

Participants will be seated at the computer and fitted with headphones. They will complete a brief 30-second practice typing task using a different passage to familiarise themselves with the keyboard and software. Participants will then receive standardised instructions: "You will hear some music through the headphones. While the music plays, please type the passage shown on screen as quickly and accurately as you can. The task will last five minutes."

The music will begin and the typing passage will appear on screen. Participants will type for exactly five minutes. At the end, the music will stop and participants will cease typing. The software will automatically record words typed per minute. Participants will be thanked, fully debriefed about the true purpose of the study, and given the opportunity to withdraw their data. Contact details for the researcher will be provided.

Draft Results

Table 1

Mean Typing Speed (Words per Minute) by Music Tempo Condition

Condition	<i>n</i>	<i>M</i>	<i>SD</i>
Fast tempo (140 BPM)	20		

Slow tempo (60 BPM)	20		
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Note. BPM = beats per minute. Values left blank; this is a research plan.

Statistical Test

The data will be analysed using an independent samples t-test. This test is appropriate because: the design is independent groups (different participants in each condition); there are exactly two conditions to compare; the dependent variable (words per minute) is continuous, ratio-level data; and with 20 participants per group, the central limit theorem suggests the sampling distribution will be approximately normal.

Prior to analysis, assumptions of normality (via Shapiro-Wilk test and visual inspection of histograms) and homogeneity of variance (via Levene's test) will be checked. If assumptions are violated, a Mann-Whitney U test will be used as the non-parametric alternative.

References

- Qwerty, A. (2025). Keyboard proficiency and occupational wellbeing: A meta-analysis. *Journal of Workplace Ergonomics*, 42(3), 215-232. <https://doi.org/10.1000/jwe.2025.0042>